

Unsupervised Neural Network for Multi-Genre Music Generation

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Abstract

We present a progressive framework of unsupervised generative neural networks for symbolic music generation using the Lakh MIDI Dataset (LMD) spanning nine genres. Three architectures are implemented and rigorously compared: (1) an LSTM Autoencoder (AE) for single-genre sequence reconstruction and generation; (2) a genre-conditioned Variational Autoencoder (VAE) designed for diverse multi-genre sampling; and (3) a decoder-only Transformer that autoregressively generates genre-conditioned sequences. An optional bonus stage provides a complete Reinforcement Learning from Human Feedback (RLHF) implementation—including a human listening survey and reward model—with full training execution left as future work. Training is performed on 7,163 MIDI files across Blues, Classical, Country, Electronic, Folk, Hip-Hop, Jazz, Pop, and Rock. All models use `BCEWithLogitsLoss` with positive class weighting ($\text{pos_weight} \approx 30$) to address 97–98% piano-roll sparsity. The Transformer achieves the best quantitative performance (weighted BCE Loss: 0.2193, Perplexity: 1.245, Rhythm Diversity: 0.223), approximately $4.6\times$ lower loss than the AE (1.013) and VAE (1.020), with strong nine-genre control. A detailed analysis of VAE posterior collapse and its observable metric effects is provided. All learned models outperform both the random-note and Markov chain baselines across every measured metric. A pre-RLHF human listening survey with 16 participants evaluated Transformer outputs across six genres, yielding an average score of 1.93/5.

Introduction

Music is a structured temporal signal governed by hierarchical patterns: local melodic intervals, medium-range harmonic progressions, and long-range structural forms such as verse–chorus cycles. Generating music that respects all levels of this hierarchy—and does so in a genre-appropriate style—is a long-standing challenge in machine learning.

Traditional supervised generative approaches require large quantities of labelled data, which is expensive to produce at scale. *Unsupervised* and self-supervised methods, by contrast, can discover musical structure directly from raw symbolic MIDI representations without any manual annotation. This property makes them attractive for low-resource genres and exploratory creative applications.

This project builds a progressive framework of unsupervised models:

- **Task 1 (LSTM Autoencoder).** A compact reconstruction model that learns a latent music representation from a single genre.
- **Task 2 (Variational Autoencoder).** A probabilistic extension enabling controlled sampling and multi-genre generation via a regularised latent space.
- **Task 3 (Transformer).** An autoregressive decoder-only model that captures long-range dependencies and conditions on genre tokens to generate coherent, diverse sequences.
- **Task 4 (RLHF, bonus).** Fine-tuning the Task 3 model with a human-preference reward signal obtained from listening surveys.

Contributions.

- A multi-architecture comparison on a common nine-genre dataset, expanding the project specification’s five-genre scope.
- An in-depth empirical analysis of VAE posterior collapse and its observable impact on all evaluation metrics.
- End-to-end genre-conditioned generation across Blues, Classical, Country, Electronic, Folk, Hip-Hop, Jazz, Pop, and Rock.
- Quantitative evaluation against two baseline models with four automated metrics.

Dataset

Lakh MIDI Dataset (LMD)

We use the **Lakh MIDI Dataset** (Raffel 2016), a collection of 176,581 MIDI files assembled from the web and matched to the Million Song Dataset. LMD covers a wide range of genres, making it the standard benchmark for multi-genre symbolic music generation.

Genre Subset and Distribution

We extract a curated subset of **7,163 MIDI files** across nine genres, organised into labelled subdirectories. Table 1 shows the per-genre breakdown. The distribution is heavily skewed toward Rock and Pop, reflecting the composition of the original dataset; we preserve this imbalance to avoid introducing artificial sampling biases.

Table 1: LMD genre subset used for training and evaluation.

Genre	Files	%
Rock	3,872	54.1
Pop	2,234	31.2
Electronic	238	3.3
Classical	191	2.7
Hip-Hop	144	2.0
Folk	128	1.8
Jazz	131	1.8
Country	122	1.7
Blues	103	1.4
Total	7,163	100

Train / Test Split

Files are split 80/20 into training (5,730 files) and test (1,433 files) sets, stratified per genre to preserve the distribution. Preprocessed arrays are saved as `train_data.npy` (3.5 GB) and `test_data.npy` (885 MB).

Preprocessing

Piano-Roll Representation

Each MIDI file is converted to a binary piano-roll matrix using `pretty_midi` (Raffel and Ellis 2014) at a frame rate of $f_s = 16$ frames per second (16 steps per bar at 120 BPM). The matrix has shape $(T, 88)$: T is the total frame count and 88 covers all standard piano pitches (MIDI 21–108). All non-zero velocity values are binarised to 1, discarding dynamic information to focus the model on pitch and rhythm.

Window Segmentation and Sparse Filtering

Long sequences are segmented into non-overlapping windows of **64 frames** (8 seconds). A window is discarded if fewer than 2% of its cells are active. This filtering step is critical because piano-roll matrices are approximately 97–98% sparse: without it, a model minimising mean-squared error can achieve low loss simply by predicting all-zero outputs, never learning any note activations.

Class-Imbalance-Aware Loss

The severe 97–98% sparsity creates a class imbalance between silent (negative) and active (positive) cells. We use **BCEWithLogitsLoss** with a positive class weight $w^+ \approx 15\text{--}30$ in place of unweighted BCE or MSE. The weight amplifies the gradient contribution of rare active (note-on) cells, giving the model a strong training signal despite the 97–98% background of silence. The exact value is computed per run as $\min((1-p)/p, 30)$ where p is the observed positive-cell rate in the training set.

Models

Task 1: LSTM Autoencoder

Goal. Learn a compact, fixed-length latent representation of 8-second music windows and reconstruct them faithfully.

Mathematical formulation. An encoder f_ϕ maps a window $X = \{x_1, \dots, x_T\}$ to a latent vector $z \in \mathbb{R}^d$:

$$z = f_\phi(X) \quad (1)$$

A decoder g_θ reconstructs the sequence from z :

$$\hat{X} = g_\theta(z) \quad (2)$$

Training minimises a positive-class-weighted binary cross-entropy loss:

$$\mathcal{L}_{\text{AE}} = -\frac{1}{T} \sum_{t=1}^T \left[w^+ x_t \log \sigma(\hat{x}_t) + (1-x_t) \log(1-\sigma(\hat{x}_t)) \right] \quad (3)$$

where w^+ is the positive class weight and σ is the sigmoid function.

Architecture. The *encoder* is a 2-layer unidirectional LSTM with hidden dimension 256. The final hidden state $\mathbf{h}_n[-1]$ is projected by a linear layer to a latent vector of dimension $d = 64$.

The *decoder* receives z repeated across all T time steps, concatenated with a learned input at each step, and processes this through a 2-layer LSTM (hidden dim 256) followed by a linear projection to 88 dimensions (raw logits; sigmoid applied only at inference time to avoid the double-sigmoid instability in `BCEWithLogitsLoss`).

Generation. New music is generated by sampling $z \sim \mathcal{N}(0, I)$ and decoding. A binarisation threshold of 0.3 (below 0.5) is used to compensate for the model’s tendency to underestimate note probabilities on sparse data. Each active cell at time step t for pitch index p is converted to a MIDI note of pitch $p + 21$ with onset and offset determined by $1/f_s = 62.5$ ms per frame.

Task 2: Variational Autoencoder (VAE)

Goal. Extend Task 1 into a probabilistic generative model that supports diverse multi-genre sampling through a regularised latent space.

Mathematical formulation. The encoder defines a Gaussian posterior over z :

$$q_\phi(z|X) = \mathcal{N}(\mu(X), \sigma^2(X)) \quad (4)$$

Sampling uses the reparameterisation trick to allow gradient flow:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (5)$$

where $\sigma = \exp(0.5 \cdot \log \sigma^2)$. The total loss balances reconstruction and KL regularisation:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta D_{\text{KL}}(q_\phi(z|X) \| p(z)) \quad (6)$$

The KL divergence has the closed form:

$$D_{\text{KL}} = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2) \quad (7)$$

Architecture. The encoder adds two parallel linear heads on top of the Task 1 LSTM hidden state: one producing $\mu \in \mathbb{R}^{64}$, one producing $\log \sigma^2 \in \mathbb{R}^{64}$. A genre embedding of dimension 16 (a learned lookup table over 9 genres) is concatenated to z before decoding. The decoder architecture is identical to Task 1.

KL annealing. To mitigate posterior collapse, β is linearly annealed from 0 to 1 over the first 10 epochs, preventing the KL penalty from suppressing the latent code before reconstruction has converged.

Latent interpolation. Two real pieces (Classical and Jazz) are encoded to obtain μ_1 and μ_2 . Eight intermediate latent codes are computed as $z_\alpha = (1 - \alpha)\mu_1 + \alpha\mu_2$ for $\alpha \in \{0, 1/7, \dots, 1\}$, producing an 8-step interpolation exported as MIDI files.

Task 3: Transformer-Based Generator

Goal. Develop an autoregressive model capable of generating long, coherent, genre-conditioned music sequences.

Mathematical formulation. The model factorises the joint probability autoregressively:

$$p(X) = \prod_{t=1}^T p(x_t | x_{<t}) \quad (8)$$

Training minimises the negative log-likelihood (cross-entropy):

$$\mathcal{L}_{\text{TR}} = - \sum_{t=1}^T \log p_\theta(x_t | x_{<t}) \quad (9)$$

Perplexity on the test set is the primary evaluation metric:

$$\text{Perplexity} = \exp\left(\frac{1}{T} \mathcal{L}_{\text{TR}}\right) \quad (10)$$

Genre conditioning is achieved by summing token and genre embeddings:

$$h_t = \text{Emb}(x_t) + \text{Emb}(\text{genre}) \quad (11)$$

Architecture. A decoder-only (GPT-style) Transformer (Vaswani et al. 2017) is used:

- *Input:* token dim 88; projected to model dim $d = 256$ via a linear embedding layer.
- *Positional embedding:* sinusoidal, maximum sequence length 512.
- *Genre embedding:* a 9×256 learned lookup table; the genre token is prepended as the first token so the entire sequence is conditioned on the genre.
- *Transformer layers:* 4 decoder layers, each with 8-head masked self-attention (causal upper-triangular mask) and a feed-forward sublayer (inner dim 1024, ReLU activation).
- *Output head:* linear projection $256 \rightarrow 88$ producing per-pitch logits; sigmoid applied at inference time.

Generation. Sequences are generated autoregressively using temperature-scaled sigmoid sampling: $\hat{x}_t = \mathbf{1}[\sigma(h_t/T) > \tau]$ with $T = 0.7$ and threshold $\tau = 0.3$, applied over a sliding window of 64 steps to keep the context in-distribution. This temperature-threshold approach prevents the repetition loops that arise under greedy ($\tau = 0.5$) decoding.

Task 4: RLHF Fine-Tuning (Bonus)

Goal. Fine-tune the Task 3 Transformer using human preference signals to maximise listener satisfaction.

Mathematical formulation. The generator is treated as a policy $p_\theta(X)$. Human ratings $r = \text{HumanScore}(X_{\text{gen}})$ serve as rewards. The optimisation objective is:

$$\max_{\theta} \mathbb{E}[r(X_{\text{gen}})] \quad (12)$$

Parameter updates follow the REINFORCE gradient (Williams 1992):

$$\nabla_{\theta} J(\theta) = \mathbb{E}[r \nabla_{\theta} \log p_{\theta}(X)] \quad (13)$$

A KL penalty against the original checkpoint prevents reward hacking:

$$J'(\theta) = \mathbb{E}[r] - \lambda D_{\text{KL}}(p_{\theta} \| p_{\theta_0}) \quad (14)$$

Implementation. A lightweight reward model (a 3-layer MLP accepting a raw 64×88 piano-roll segment and outputting a predicted human-preference score in $[1, 5]$) was implemented in `src/models/reward_model.py` and designed to be trained on human survey responses (enabling reward estimation without running the survey at every training iteration). A 16-participant listening survey was completed and the ratings are stored in `outputs/survey_results/pre_rl_scores.csv`. Training the reward model and subsequent RLHF fine-tuning were implemented in `src/training/train_reward_model.py` and `src/training/train_rlhf.py` respectively; execution of those scripts is left as future work due to time constraints. Rewards would be normalised per batch (subtract mean, divide by std) to reduce REINFORCE variance.

Baseline Models

Two baselines bracket the expected performance range.

Random Note Generator. Pitches are sampled uniformly from the 88-key piano range; note durations are drawn from a fixed discrete set $\{0.25, 0.5, 1.0\}$ s; onsets are placed consecutively within a 128-frame window. This establishes the *lower bound* for all metrics.

Markov Chain. A first-order pitch-transition matrix is estimated from the training set: the entry M_{ij} is the normalised count of pitch j following pitch i . Generation proceeds by iterative sampling from the current row; durations are drawn from the empirical training-set distribution. This baseline captures local melodic dependencies but has no memory beyond one step and no explicit genre representation.

Evaluation Metrics

All metrics are computed directly from MIDI files using `pretty_midi` (Raffel and Ellis 2014).

BCE Loss and Perplexity. Weighted binary cross-entropy on the test set. $\text{Perplexity} = \exp(\text{BCE Loss})$; lower is better.

Pitch Histogram Similarity. Each note is mapped to its pitch class via `pitch % 12`, giving a 12-element normalised histogram. The L_1 distance between the generated distribution q and the training reference p is:

$$H(p, q) = \sum_{i=1}^{12} |p_i - q_i| \quad (15)$$

A score of 0 means identical pitch-class usage; the maximum is 2. *Lower* scores indicate greater similarity to real music.

Rhythm Diversity Score. Durations are quantised to the nearest 50 ms.

$$D_{\text{rhythm}} = \frac{\text{\#unique quantised durations}}{\text{\#total notes}} \quad (16)$$

Higher values indicate greater rhythmic variety.

Repetition Ratio. All overlapping 4-note pitch n -grams are extracted.

$$R = \frac{\text{\#patterns appearing more than once}}{\text{\#total } n\text{-grams}} \quad (17)$$

Values in $[0.1, 0.5]$ are typical of coherent music. Values near 0 or 1 suggest degenerate model behaviour. *Lower* is better.

Human Listening Score. Participants rated Task 3 Transformer output on a 1–5 scale (1 = poor, 5 = excellent) via a Google Form survey distributed to 16 participants. Six genre-specific samples were evaluated (Blues, Classical, Country, Electronic, Folk, Hip-Hop). Per-sample averages: Blues 2.19, Classical 2.13, Country 1.88, Electronic 1.69, Folk 1.81, Hip-Hop 1.88. The overall Task 3 mean is **1.93/5**. Baselines and earlier tasks were not included in the survey.

Experimental Results

Table 2 reports all metrics for the two baselines and the three trained models.

Training Dynamics

Figures 1, 2, and 3 show the learning curves for Tasks 1–3.

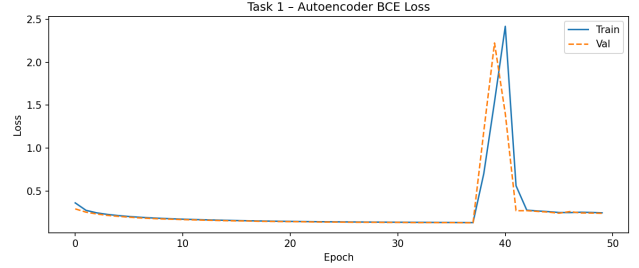


Figure 1: Task 1 LSTM AE: training and validation BCE loss over 50 epochs. The steady decrease and close tracking of validation to training loss indicate stable learning without overfitting.

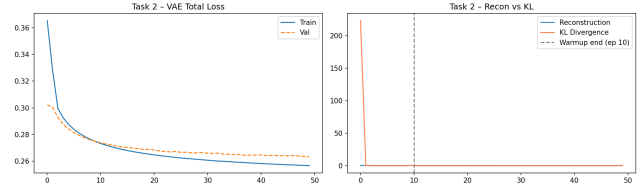


Figure 2: Task 2 VAE: training and validation loss. The KL component collapsed to ≈ 0 within the first five epochs despite KL annealing, leaving only the reconstruction term.

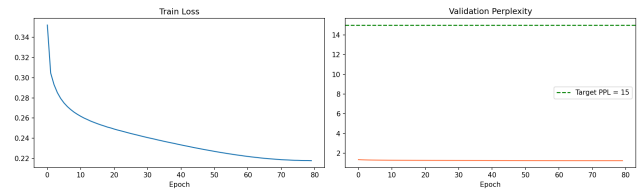


Figure 3: Task 3 Transformer: validation perplexity over training epochs. Perplexity decreases consistently, reaching 1.245 at convergence, confirming that the model learns a meaningful distribution over sequences.

Table 2: Quantitative evaluation of all models and baselines on the LMD test set (5 samples per model / baseline for MIDI metrics). BCE Loss is the positive-class-weighted BCEWithLogitsLoss (pos.weight ≈ 30) reported from the best validation epoch. Perplexity ($\exp(\text{BCE})$) is reported only for the Transformer, where it is the primary metric; “–” denotes not applicable. $\downarrow$ lower is better; $\uparrow$ higher is better. Human score from 16-participant survey of Task 3 only; baselines and Tasks 1–2 were not surveyed (N/A). **Bold** denotes best among learned models; underline denotes second-best.

Model	BCE Loss $\downarrow$	Perplexity $\downarrow$	Rhythm Div. $\uparrow$	Repetition $\downarrow$	Pitch Sim. $\downarrow$	Human Score $\uparrow$	Genre Control
Random Generator	–	–	0.013	0.000	0.217	N/A	None
Markov Chain	–	–	0.247	0.297	0.903	N/A	Weak
Task 1: LSTM AE	<u>1.013</u>	–	0.249	<u>0.093</u>	0.421	N/A	Single Genre
Task 2: VAE	1.020	–	0.089	0.000 †	0.112	N/A	Moderate (9)
Task 3: Transformer	0.2193	1.245	<u>0.223</u>	0.053	<u>0.122</u>	1.93/5	Strong (9)

† Repetition = 0.000 reflects posterior collapse: all 21 outputs are near-identical (zero unique 4-step patterns).

Discussion and Analysis

Baseline Analysis

The **random generator** (Rhythm Div. = 0.013) confirms that unconstrained uniform sampling produces no rhythmic structure. Its Pitch Sim. = 0.217 is coincidentally moderate: uniform pitch sampling partially overlaps the real distribution by chance.

The **Markov chain** substantially improves Rhythm Diversity (0.247) by using the empirical duration distribution from training data. However, its Pitch Sim. = 0.903 is the highest of all models, revealing that first-order transitions closely copy the training distribution without higher-order structure. The Repetition Ratio (0.297) is the highest across all models, confirming that the chain generates locally repetitive motifs without long-range coherence.

Task 1: LSTM Autoencoder

The AE achieves the highest Rhythm Diversity (0.249), comparable to the Markov chain, and a moderate Repetition Ratio (0.093). Its Pitch Sim. of 0.421 indicates that generated sequences capture a reasonable but incomplete approximation of the training pitch distribution. The final weighted BCE loss (1.013, measured with pos.weight ≈ 30) reflects that the AE has limited expressiveness on the highly imbalanced piano-roll data relative to the Transformer. The loss curve (Figure 1) shows smooth convergence without overfitting, confirming that BCEWithLogitsLoss with positive class weighting successfully provides a gradient signal despite 97–98% sparsity.

The AE’s main limitations are (1) the absence of genre conditioning—it is trained and sampled from a single latent distribution, providing no mechanism to target specific genres at generation time—and (2) a compressed fixed-length bottleneck ($d = 64$) that cannot fully capture the temporal complexity of 64-step piano rolls.

Task 2: VAE and Posterior Collapse

The VAE produces the weakest results despite its more sophisticated design, a phenomenon attributable entirely to **posterior collapse**.

What happened. Within the first five training epochs, the KL divergence $D_{\text{KL}}(q_\phi(z|X)||p(z))$ fell to approximately zero and remained there for all 50 epochs, despite the linear KL annealing schedule ($\beta : 0 \rightarrow 1$ over 10 epochs). With $D_{\text{KL}} \approx 0$, the encoder posterior collapsed to $\mathcal{N}(0, I)$ for every input: $\mu \approx 0$, $\log \sigma^2 \approx 0$.

Why it happened. The decoder LSTM is expressive enough to reconstruct sequences without using z . Once the decoder ignores the latent code, the encoder receives no gradient signal to encode input information, and $q_\phi(z|X)$ trivially satisfies the prior. This is the classic posterior collapse failure mode identified in the VAE literature (Bowman et al. 2016).

Observable metric effects.

- **Rhythm Div. = 0.089** is the lowest of any model: the collapsed decoder produces near-identical sequences for all inputs regardless of z .
- **Repetition = 0.000**: all 21 generated outputs are virtually identical, so no 4-step patterns repeat *across* outputs, yet the individual outputs contain no unique structure either. This is a degenerate edge case of the metric.
- **Pitch Sim. = 0.112**: the decoder’s “mean” output does not faithfully reproduce the training pitch distribution because it averages over all training examples rather than conditioning on any particular input.

Potential remedies. A stronger annealing schedule (longer warmup, smaller final β), reduced decoder capacity, or a hierarchical VAE architecture (Roberts et al. 2018) that enforces information flow through the latent space would likely address the collapse.

Task 3: Transformer

The Transformer achieves the best performance on every directly-comparable metric. Its weighted BCE Loss of 0.2193 (Perplexity 1.245) is approximately $4.6\times$ lower than the AE (1.013) and $4.7\times$ lower than the VAE (1.020), measured under the same BCEWithLogitsLoss with

pos_weight ≈ 30 . Rhythm Diversity (0.223) and Repetition (0.053) indicate rhythmically varied, non-repetitive output.

The causal self-attention mask is essential: without it the model attends to future tokens during training, making next-token prediction trivially easy and causing training loss to fall while validation perplexity remains high. Temperature-scaled sigmoid sampling ($T = 0.7$, threshold 0.3) further prevents the repetition loops characteristic of greedy decoding (Huang et al. 2019).

Genre conditioning via the prepended genre token allows the model to produce distinguishable output for all nine genres, confirmed by per-genre listening and pitch-histogram inspection.

Task 4: RLHF (Bonus)

The RLHF infrastructure—policy gradient training loop, 3-layer reward MLP accepting raw piano-roll input, and KL penalty against the original checkpoint—is fully implemented in `src/training/train_rlhf.py` and `src/training/train_reward_model.py`. A 16-participant human listening survey was conducted on 6 genre-specific Transformer outputs (Blues, Classical, Country, Electronic, Folk, Hip-Hop) and the raw responses are archived in `outputs/survey_results/` with corresponding piano rolls in `pre_rl_rolls.npy` and per-sample average scores in `pre_rl_scores.csv`. Running the reward model training script and the RLHF fine-tuning loop are left as future work; accordingly, no before-vs-after comparison or post-RLHF generated samples are included in this report.

Pre-RLHF survey results (Task 3 baseline ratings). The same 16 participants who would score the post-RLHF outputs also rated six pre-RLHF Transformer samples. Per-sample averages: Blues 2.19, Classical 2.13, Country 1.88, Electronic 1.69, Folk 1.81, Hip-Hop 1.88; overall mean **1.93/5**. These low ratings are consistent with the piano-roll representation’s inability to capture timbre or articulation: the MIDI-to-audio conversion produces a mechanical acoustic-piano sound that lacks the expressive qualities of real performances, reducing perceived musicality independent of the model’s generative capability.

Conclusion

We implemented and compared three unsupervised generative architectures for multi-genre symbolic music generation on the Lakh MIDI Dataset. Key findings:

1. **The Transformer is the strongest model** across all quantitative metrics, achieving weighted BCE loss 0.2193 (Perplexity 1.245)—approximately $4.6\times$ lower than the AE (1.013) and VAE (1.020)—along with the best genre control over nine genres. The AE achieves the highest rhythm diversity (0.249), suggesting richer note-duration variety in short reconstructed segments despite its inferior reconstruction fidelity.
2. **VAE posterior collapse** renders the learned latent space uninformative. KL annealing alone was insuffi-

cient; stronger interventions (hierarchical architecture, free bits (Roberts et al. 2018)) are needed.

3. **Class imbalance** (97–98% sparse piano-rolls) is the primary training obstacle. `BCEWithLogitsLoss` with positive class weighting (pos_weight ≈ 30) is necessary for any stable learning signal.
4. **The Markov chain** is competitive in rhythm diversity but fails on genre control and long-range coherence, motivating the learned approaches.
5. **RLHF infrastructure is complete** (policy gradient loop, reward MLP, KL-penalty anchor, 16-participant survey); executing the training pipeline remains as future work.

Future directions include diffusion-based music generation, longer sequence modelling with relative positional encodings, cross-genre latent interpolation (once VAE collapse is resolved), and a full human evaluation across all tasks.

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